

Claude AI Learning Roadmap for AWS & DevOps Career Growth

This roadmap is designed for your background:

- AWS Infrastructure
- Terraform
- GitHub & GitHub Actions
- CI/CD
- Beginner in coding and scripting

Goal: Learn Claude AI in a practical way for Cloud, DevOps, Automation, and Career Growth.

Phase 2: Learn Prompt Engineering (Week 2-3)

Goal: Learn how to ask Claude AI correctly to get accurate, professional, and useful answers.

Prompt Engineering = Writing good instructions for AI.

Good prompt = Better answer.

Week 2: Learn Prompt Basics

Day 1: Understand What Prompt Engineering Means

Definition

Prompt engineering means:

- Asking AI clearly
- Giving context
- Defining output format
- Getting accurate response

Example

Bad Prompt:

"AWS"

Good Prompt:

"Explain AWS VPC architecture for beginner with real-time example."

Important Rule

More details = Better answer

Day 2: Learn Prompt Formula

Best Prompt Formula

Prompt = Role + Task + Context + Output

Example

Role: "Act as AWS architect"

Task: "Create Terraform code"

Context: "Need VPC, EC2, ALB"

Output: "Provide step-by-step explanation"

Final Prompt Example

"Act as AWS architect. Create Terraform code for VPC, EC2, and ALB. Use AWS best practices. Explain step-by-step."

Day 3: Learn Simple Prompt Types

1. Explain Prompt

Used for learning.

Examples:

- Explain AWS subnet
- Explain Terraform module
- Explain GitHub workflow

2. Generate Prompt

Used for creating things.

Examples:

- Generate Terraform code
- Generate YAML file
- Generate Dockerfile

3. Debug Prompt

Used for fixing issues.

Examples:

- Fix Terraform error
- Fix AWS issue
- Fix YAML problem

4. Compare Prompt

Used for decision making.

Examples:

- Compare ECS vs EKS
- Compare EC2 vs Lambda
- Compare Terraform vs CloudFormation

Day 4: Learn Good Prompt vs Bad Prompt

Bad Prompt Example

"Create VPC"

Good Prompt Example

"Create Terraform code for AWS VPC with:

- 2 public subnet
- 2 private subnet
- Internet Gateway
- NAT Gateway
- Route table Explain every section."

Key Difference

Good prompt gives:

- Clear requirements
 - Better output
 - Less confusion
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Day 5: Learn Follow-up Prompting

Claude Remembers Context

You can continue conversation.

Example

Prompt 1:

"Create AWS architecture for website."

Prompt 2:

"Add ALB to architecture."

Prompt 3:

"Add Auto Scaling also."

Prompt 4:

"Convert to Terraform code."

Practice

Create one long conversation.

Day 6: Learn Output Control

Control Claude Output

You can ask Claude to give output in specific format.

Examples:

- Table format
- Step-by-step format
- Beginner explanation
- Professional explanation
- Architecture diagram explanation

Example Prompt

"Explain AWS VPC in table format with examples."

Day 7: Week 2 Practice

Practice Task

Create prompts for:

- AWS VPC
- Terraform
- GitHub Actions
- Kubernetes
- Docker

Goal

Practice asking better questions.

Week 3: Advanced Prompt Engineering

Day 8: Learn Role-Based Prompting

Role Prompt

Tell Claude who it should act like.

Examples:

- Act as AWS Solution Architect
- Act as DevOps Engineer
- Act as Terraform Expert
- Act as Python Trainer

Example

"Act as AWS architect. Design secure VPC architecture."

Day 9: Learn Step-by-Step Prompting

Use for Learning

Example:

"Teach Terraform from beginner to expert with examples."

Use for Troubleshooting

"Fix AWS ALB issue step-by-step."

Day 10: Learn Constraint Prompting

Add Rules to Claude

Examples:

- Beginner level explanation
- No coding knowledge
- Simple English
- AWS best practices
- Low-cost architecture

Example

"Explain EKS in simple language for non-programmer."

Day 11: Learn Real DevOps Prompting

Terraform Prompt

"Create Terraform code for ALB with EC2 and Auto Scaling."

GitHub Actions Prompt

"Create GitHub Actions pipeline to deploy Terraform code."

AWS Prompt

"Create secure AWS 3-tier architecture."

Day 12: Learn Troubleshooting Prompt

Example

"My EKS pod gives CrashLoopBackOff. Provide troubleshooting steps."

Example

"My Terraform apply fails. Help debug issue."

Day 13: Learn Documentation Prompt

Examples

"Create AWS architecture documentation."

"Write Terraform project documentation."

"Generate KT document for AWS migration."

Day 14: Week 3 Mini Project

Project

Use Claude to create full DevOps solution.

Ask Claude

1. Design AWS architecture
 2. Generate Terraform code
 3. Create GitHub Actions pipeline
 4. Add monitoring
 5. Add security best practice
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Phase 2 Final Outcome

After Week 2–3 you should know:

- How to ask Claude correctly
 - How to get better answers
 - How to create professional prompts
 - How to troubleshoot using prompts
 - How to generate AWS and DevOps solutions
 - How to use Claude like assistant engineer
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